



Failure Analysis & Reliability

Why hardware fails — and how to predict it

HALT • HASS • MTBF • Arrhenius • Bathtub curve

Audience: Intermediate Engineering Students

Failure modes • Accelerated tests • Statistical reliability • RCA



Why Reliability Engineering Matters

Field failures = real cost

Cost

Each field failure costs
10-100× a factory defect.

Brand damage

Reviews + word-of-mouth
build or break products.

Warranty

Direct dollar cost
of returns + replacement.

Safety

Some failures kill —
medical, automotive, aerospace.

Specification

Customers demand MTBF
numbers in B2B contracts.

Improvement

Analyse failures →
improve next-gen design.

The Bathtub Curve

Failure rate over product life

- ▶ Infant mortality — early failures (manufacturing defects, weak components)
- ▶ Useful life — low constant failure rate (chance failures)
- ▶ Wear-out — increasing rate (electrolytic dry-out, mechanical wear)
- ▶ Burn-in catches infant mortality at the factory
- ▶ Useful life is what you design for in MTBF specs
- ▶ Maintenance / refresh extends or replaces wear-out parts
- ▶ Different chemistries have different curves: Li-ion < electrolytic cap < FR-4

Common Failure Modes

What actually breaks in the field

Solder joint

Cracks from thermal cycling, vibration.

Electrolytic cap

Dries out, ESR rises, ripple kills regulator.

Connector

Corrosion, insertion-wear, intermittent contact.

PCB delamination

Multilayer separates from heat or moisture.

Tin whisker

Lead-free Sn grows whiskers
→ short circuit.

ESD damage

Gradual or sudden IC degradation.

Electromigration

Atoms move in metals

Corrosion

Moisture + chemistry +

Mechanical fatigue

Vibration breaks

Reliability Metrics

MTBF, MTTF, FIT, AFR — what they mean

Metric	Definition	Use
MTBF	Mean Time Between Failures (repairable system)	Servers, industrial
MTTF	Mean Time To Failure (non-repairable)	Disposable / wear items
FIT	Failures In Time = failures per 10^9 hours	Component-level rating
AFR	Annualised Failure Rate (%)	Consumer warranty calc
MTTR	Mean Time To Repair	Service KPI
Reliability	$P(\text{no failure in } T) = e^{(-\lambda t)}$	Statistical analysis
Cpk	Process capability (manufacturing)	Production quality

Example MTBF claim

10^6 hours MTBF = 114 years — meaningless for one device. Statistical for fleet: out of 10000 units, ~1% will fail per year.

Arrhenius — Temperature Acceleration

Heat is the universal aging accelerator

- ▶ Arrhenius equation: failure rate $\propto e^{(-E_a/kT)}$
- ▶ Activation energy E_a (eV) — chemistry-dependent
- ▶ Rule of thumb: every +10 °C halves life ($E_a \approx 0.7$ eV)
- ▶ Electrolytic caps: rated life $\times 2^{((105 - T_{\text{actual}})/10)}$
- ▶ Aluminum cap rated 2000 h at 105 °C = ~16000 h at 75 °C
- ▶ Run-hot products age fast — keep junction temp $\leq 85\%$ of $T_{J\text{max}}$
- ▶ Acceleration factor used in HALT/HASS to compress years into days

HALT vs HASS

Accelerated stress testing

HALT — Highly Accelerated Life Test

- ▶ Pre-production / design phase
- ▶ Stress until failure — find weakest links
- ▶ Temperature -40 to +85 °C cycling
- ▶ Vibration random 5-50 G
- ▶ Combined temp + vibe at the same time
- ▶ Result: design improvements, MTBF model

HASS — Highly Accelerated Stress Screen

- ▶ Production phase — screening
- ▶ Stress within margins to catch defects
- ▶ Reduces infant mortality in shipped products
- ▶ Faster than burn-in
- ▶ All shipped units pass HASS
- ▶ Result: lower field-failure rate

Other Accelerated Tests

Specific environmental stresses

Thermal cycle

-40 to +85 °C × 1000.
Catches solder cracks.

Damp heat

85 °C / 85% RH for 1000h.
Finds corrosion.

Vibration

Random + sine sweep.
Mechanical robustness.

Mechanical shock

Drop test, transport.
For portable products.

Salt fog

1000 h salt spray.
Marine / outdoor.

ESD test

8 kV / 15 kV per IEC 61000-4-2.
ESD weak points.

Root Cause Analysis

When a board fails, dig deep

- ▶ Capture failure data — temperature, voltage, time, environment
- ▶ Photograph the unit before any disassembly
- ▶ Visual inspection under microscope — find obvious damage
- ▶ X-ray for hidden defects (BGA, solder voids)
- ▶ Decap (chemical etch) of failed ICs — die-level inspection
- ▶ Electrical characterisation — what's open, shorted, or out of spec
- ▶ Compare with known-good unit — isolate the difference
- ▶ Five Why's analysis — symptom → cause → root cause

Failure Analysis Tools

Equipment in an FA lab

Optical microscope

10-200× visual inspection.
₹15-50k.

X-ray inspection

See inside BGA, find voids.
₹15-50 L.

SEM / EDX

Element-level imaging.
₹50 L-2 Cr.

Decapsulator

Chemical etch to expose die.
Lab service.

Thermal imager

Hotspots, power leak.
₹20-200k.

Curve tracer

Component I-V characterisation.
₹10-50 L.

Design for Reliability (DfR)

Habits that boost MTBF

- ▶ Derate all components (70% rule)
- ▶ Avoid borderline tolerances — design for worst case
- ▶ Multiple grounds, decoupling caps everywhere
- ▶ Protect every external interface (TVS, ESD)
- ▶ Thermal pads + copper pour for hot ICs
- ▶ Avoid mechanical stress on solder joints (rigid mounts only at edges)
- ▶ Use cap chemistries with long life (polymer Ta vs aluminum)
- ▶ Conformal coating for moisture & dust protection
- ▶ Vibration-tolerant connectors with positive lock
- ▶ Software watchdog — automatic recovery from glitches
- ▶ OTA firmware — fix bugs after deployment
- ▶ Statistical sampling of production — catch drift

Reliability Prediction

Estimate MTBF before shipping

- ▶ MIL-HDBK-217 — classic component reliability handbook (US)
- ▶ Telcordia SR-332 — telecom equivalent
- ▶ FIDES / IEC 62380 — modern European method (more realistic)
- ▶ Sum component FIT rates $\rightarrow$ system FIT $\rightarrow$ $MTBF = 10^9 / FIT$
- ▶ FIT rates from vendor datasheets (e.g. TI publishes FIT per IC)
- ▶ Account for temperature (Arrhenius) — published rates assume 40 °C
- ▶ Watch: handbook predictions tend to underestimate by 3-10× (real-world)
- ▶ Validate predictions with HALT data on prototype

Field Reliability Data

What customers actually experience

- ▶ RMA (Return Material Authorisation) — track every returned unit
- ▶ 8D (Eight Disciplines) Problem Solving — auto industry standard
- ▶ Pareto chart: 80% of failures come from 20% of root causes
- ▶ FRACAS (Failure Reporting, Analysis, and Corrective Action System)
- ▶ Field telemetry: log temperature, voltages, error counts
- ▶ Predictive maintenance with sensor data — preempt failure
- ▶ DOE (Design of Experiments) — find sensitivities in design
- ▶ Continuous improvement loop: data → analysis → design fix → re-launch

Common Reliability Mistakes

What field returns teach

- ▶ Skipping HALT — find weak link in field, not lab
- ▶ Underestimating actual operating temperature
- ▶ Electrolytic cap as decoupling — lifetime issue
- ▶ No conformal coating in humid / dusty environments
- ▶ Single-vendor components — supply break = redesign
- ▶ Mechanical stress on PCB-mount connectors — solder joints crack
- ▶ Tin-only finish without nickel barrier — tin whiskers
- ▶ Ignoring derating — wear-out accelerated 10-100×

Key Takeaways

Key takeaways from this lecture

Bathtub curve

Infant → useful life → wear-out.

Heat kills

Every +10 °C halves life.

HALT finds weak spots

Stress until failure.

HASS screens production

Every unit before ship.

RCA every return

Five Whys, root cause.

Predict + measure

Calc MTBF, validate in lab.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ MIL-HDBK-217F — Component Reliability handbook
- ▶ Telcordia SR-332 — telecom reliability
- ▶ FIDES Guide — modern reliability prediction
- ▶ Patrick O'Connor — 'Practical Reliability Engineering'
- ▶ Mike Silverman — 'Forensic FA' (engineering podcast)
- ▶ Reliability Hotwire — DfR Solutions newsletter (free)
- ▶ Local FA labs: SAMEER, Centre for Reliability Bangalore
- ▶ Cypress / Infineon / TI vendor reliability handbooks

Thank you • Build something this week.